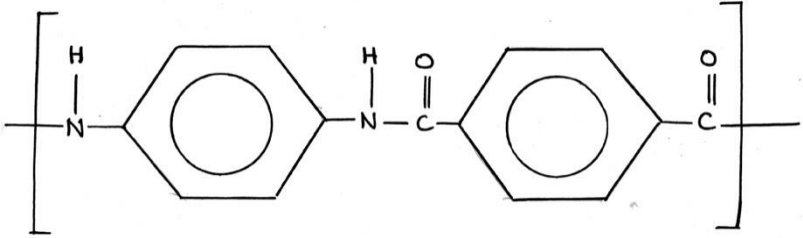
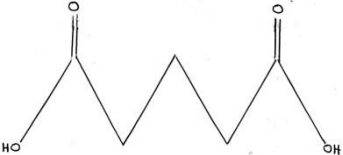


Question				Marking details	Marks available					
					AO1	AO2	AO3	Maths	Prac	Total
9.	(a)	(i)		hydrolysis	1					1
		(ii)		moles of NaOH = $\frac{75.00}{1000} \times 0.400 = 0.0300$ (1) total number of moles of acid is 0.0300 there are equal number of moles of each acid moles of R—COOH is 0.0150 $\Rightarrow$ same moles of R—COCl (1) M_r of R—COCl = $\frac{1.598}{0.0150} = 106.5$ (1)		2	1	2		3
		(iii)		award (1) for either of following • M_r of the COCl group is 63.5 therefore M_r of the R group is 43 • likely to be C ₃ H ₇ – the NMR peak area ratio 6:1 suggests (CH ₃) ₂ CH formula of the acid chloride is (CH ₃) ₂ CHCOCl (1)			2			2
	(b)	(i)		HOOC-C ₆ H ₄ -COOH + 2PCl ₅ $\rightarrow$ ClOC-C ₆ H ₄ -COCl + 2POCl ₃ + 2HCl			1			1
		(ii)				1				1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Maths	Prac	Total
	(c)	(i)		award (1) for any of following hydrochloric acid (in the presence of zinc chloride) phosphorus pentachloride phosphorus trichloride thionyl chloride accept correct formulae		1			1	1
		(ii)		ammonia accept NH ₃		1				1
		(iii)		nucleophilic substitution	1					1
		(iv)					1			1
	(d)			M_r benzene-1,4-dicarboxylic acid is 166 (1) M_r 'repeating unit' is 192 (1) mass of acid = $75 \times \frac{166}{192} \times \frac{90}{100} = 58.4 \text{ kg}$ (1)		1 1 1		1		3

Question				Marking details	Marks available					
					AO1	AO2	AO3	Maths	Prac	Total
				alternative method moles of 'repeating unit' = $\frac{75000}{192} = 390.5 \text{ mol}$ (1) moles of acid = $\frac{90}{100} \times 390.5 = 351.5$ (1) mass of acid = $\frac{351.5 \times 166}{1000} = 58.3$ (1)						
				Total question 9	2	8	5	3	1	15